

SUBMITTAL

Project

Kerr Stock Equipment

Date

Tuesday, July 18, 1905

Designer

applied@kerrcontrols.ca

Notes

KERR STOCK EQUIPMENT

Kerr Applied

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Prepared By: Kerr Applied 09:00AM

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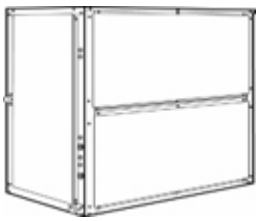
38AUDT12A0M1-0A0A0+40RFAA12A2A1-UA0A0

Submittal Cover Sheet
Unit Report
Performance Summary Report
Acoustic Summary
Certified Drawings
Guide Specifications
Feature Sheet

Unit Report For 38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:00AM





Outdoor Unit Parameters

Unit Quantity: **1**
Unit Model: **38AUD**
Unit Size: **10 Tons**
Voltage: **575-3-60** V-Ph-Hz
Condenser Coil: **Al/Cu with Louvered Hail Guard**
No. of DX Circuits: **2**
No. of Stages: **Three Stage**

Indoor Unit Parameters

Unit Model: **40RFA**
Unit Size: **10 Tons**
No. of Coils: **2**
Voltage: **575-3-60** V-Ph-Hz

System Parameter

System Quantity: **1**
Refrigerant Type: **PURON**
Compressor Quantity: **2**
Compressor Type: **Scroll**
Std. Capacity Steps: **33,83, 100**
No. of Outdoor fans: **2**

Outdoor Unit Dimensions and Weight

Unit Length: **4' 11.4"**
Unit Width: **3' 9.9"**
Unit Height: **4' 2.4"**
Unit Weight: **516** lb

Indoor Unit Dimensions and Weight

Unit Length: **4' 1.0"**
Unit Width: **2' 4.2"**
Unit Height: **4' 8.1"**
Unit Weight: **425** lb

Warranty Information Outdoor (Note: for US & Canada only)

First Year - Parts Only (Standard)

Warranty Information Indoor (Note: for US & Canada only)

First Year - Parts Only (Standard)

NOTE: Please see Warranty Catalog 808-218 for explanation of policies and ordering methods.

Ordering Information

Part Number	Description	Quantity
Base Unit - Outdoor		
38AUDT12A0M1-0A0A0		1
	Base Unit	
	Al/Cu with Louvered Hail Guard Condensing Coil	1
	Dual Circuit Three Cooling Stages Refrigerant Options	1
	Service Options - None	1
	Electrical Options - None	1
	Packaging Options - Standard	1
	Standard Electrical Mechanical Controls	1
	Refrig Circ/Compressor Staging - Dual Circuits/ Three Stage	1
Base Unit - Indoor		
40RFAA12A2A1-UA0A0		1
	Base Unit	

	Std Static, Std Effy Motor/ Medium Drive	1
	Cabinet Paint - None	1
	2-Speed Blower (VAF)	1
	Electromechanical Unit Control Board	1

The 38AUZ/AUD are furnished with filter drier which is factory provided (field installed). Additional filter driers can be purchased separately through RCD (Replacement Components Division). See the Product Library for Replacement Filter Drier Information for more information.

System:	**38AUD012/40RFA_12**	
No. of Circuits:	**Dual Circuit**	
System Quantity:	**1**	
Altitude:	**0.0**	ft
EER @ ARI Conditions:	**11.2**	
IEER:	**14.9**	
Suction Line Loss:	**1.4**	°F

System meets ASHRAE 90.1-2023 min IEER requirement.

This system has a combined part load efficiency (IEER) that meets or exceeds the US DOE 2023 minimum cooling efficiency standards.

Refer to the 38AU installation instructions for installation specific line sizing guidance

Outdoor Unit Parameters

Unit Quantity:	**1**	
PartNumber:	**38AUDT12A0M1-0A0A0**	
Unit Model:	**38AUD**	
Unit Size:	**10 Tons**	
Condenser Coil:	**Al/Cu with Louvered Hail Guard**	
No. of DX Circuits:	**2**	
Voltage:	**575-3-60**	V-Ph-Hz
Clg Ent Air DB:	**95.0**	°F

Outdoor Electrical Data

Unit Voltage:	**575-3-60**	V-Ph-Hz
Unit MCA:	**16.0**	Amps
Unit MOCP:	**20.0**	Amps
Total Compressor Power of Unit:	**NA**	
Voltage Range Min:	**518**	V
Voltage Range Max:	**633**	V
Compressor RLA:	**6.4/5.8**	
Compressor LRA:	**48/39**	
Compressor Quantity:	**2**	
Fan Motors Qty:	**2**	
Fan Motor FLA:	**0.7**	Amps

Notice: Outdoor unit elect. data is based on 575-3-60
Control Panel SCCR: 5kA RMS at Rated Symmetrical Voltage

Control Panel SCCR: 5kA RMS at Rated Symmetrical Voltage

Indoor Unit Parameters

PartNumber:	**40RFAA12A2A1-UA0A0**	
Unit Model:	**40RFA**	
Unit Size:	**10 Tons**	
Coil Rows:	**4**	
No. of Splits:	**2**	
Voltage:	**575-3-60**	V-Ph-Hz
Actual Airflow:	**3000.0**	CFM
Standard Airflow:	**3000.0**	CFM
Total Clg Cap.(Gross):	**113.6**	MBH
Sensible Clg Cap.(Gross):	**75.3**	MBH
Ent Air DB:	**80.0**	°F
Ent Air WB:	**67.0**	°F
Ent Enthalpy:	**31.44**	BTU/lb
Lvg Air DB:	**56.8**	°F
Lvg Air WB:	**54.8**	°F
Lvg Enthalpy:	**23.02**	BTU/lb
Coil Bypass Factor:	**NA**	

Indoor Supply Fan

Indoor Unit External Static:	**0.60**	in wg
Economizer Loss:	**0.00**	in wg
Grille Loss:	**0.00**	in wg
Plenum Loss:	**0.00**	in wg
Dehumidification Loss:	**0.00**	in wg
Acc. Heating Loss:	**0.00**	in wg
Total Ext Static:	**0.60**	in wg
Fan Speed:	**1271**	RPM
Fan Power:	**0.72**	BHP
Fan Motor Max:	**2.40**	BHP
Fan Motor FLA:	**2.5**	Amps

Motor and Medium Static Drive Required.

Indoor Electrical Data

Unit Voltage:	**575-3-60**	V-Ph-Hz
Unit MCA:	**4.0**	Amps
Unit MOCP:	**15.0**	Amps

Notice: Indoor unit elect. data is based on 575-3-60

Acoustics

Sound Power Levels, db re 10E-12 Watts

A-Weighted	Outdoor Unit (dB)	Indoor Unit (dB,Ducted)
Total Level	82.7	91.7
63Hz	36.7	72.4
125Hz	53.5	78.5
250Hz	65.8	82.0
500Hz	74.7	86.4
1000Hz	79.3	85.6
2000Hz	77.3	84.8
4000Hz	71.7	80.6
8000Hz	60.0	NA
Sound Message	Sound for AUD_12	Sound for RFA_12

Acoustic Notes:

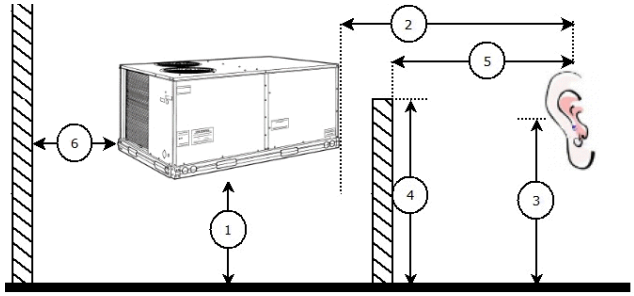
1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. 40RL/40RF/40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
3. The acoustic center of the unit is located at the geometric center of the unit.

Performance Summary For 38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:00AM

Advanced Acoustics Parameters

1. Unit height above ground: **1.0** ft
2. Horizontal distance from unit to receiver: **20.0** ft
3. Receiver height above ground: **5.7** ft
4. Height of obstruction: **0.0** ft
5. Horizontal dist. from obstruction to receiver: **0.0** ft
6. Horizontal dist. from unit to obstruction: **0.0** ft



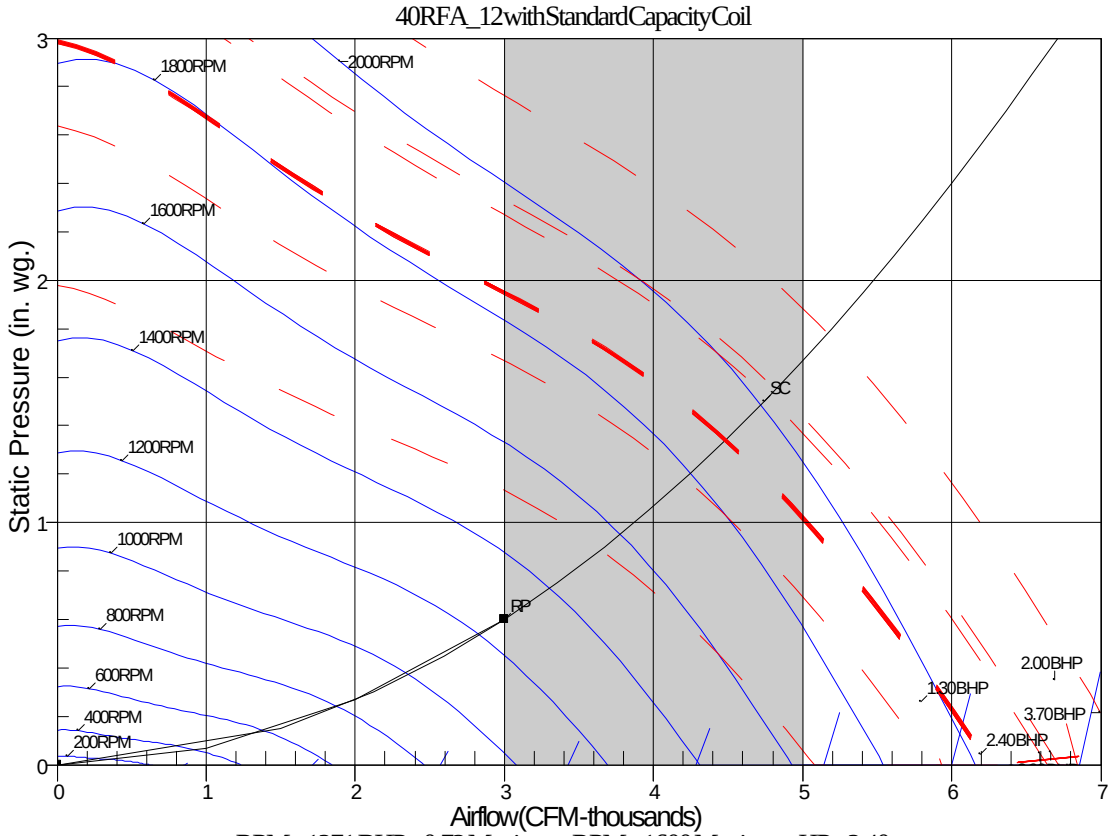
Detailed Acoustics Information

Octave Band Center Frequency, Hz	31	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	84	63	70	74	78	79	76	71	61	87
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	44	37	54	66	75	79	77	72	60	83
Sound Press. Levels at Dist. Specified above (Lp), dB	59	39	45	50	54	55	52	46	37	62
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	20	12	29	41	50	55	53	47	36	58

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Performance Summary For 38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0

Project: Kerr Stock Equipment 10/25/2026
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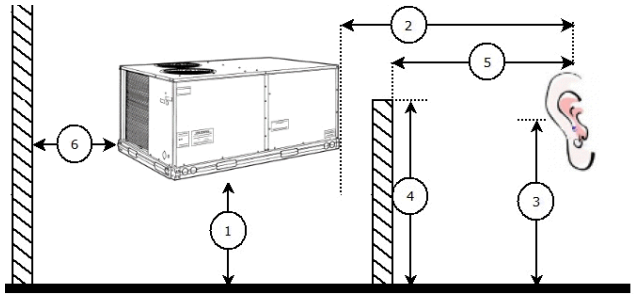
RPM=1271 BHP=0.72 MaximumRPM=1600 MaximumHP=2.40
ShadedArea-RecommendedOperatingRange
SC-SystemCurve RP-RatedPoint MP-MinimumPoint

Acoustic Summary For 38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:00AM

Outdoor Unit Parameters:

Tag Name: **38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0**
Unit Model: **38AUD**
Unit Size: **10 Tons**
System Type: **Dx Cooling Only**
Refrigerant Type: **PURON**
Compressor Quantity: **2**
Compressor Type: **Scroll**



Advanced Accoustics Parameters

1. Unit height above ground: **1.0** ft
2. Horizontal distance from unit to receiver: **20.0** ft
3. Receiver height above ground: **5.7** ft
4. Height of obstruction: **0.0** ft
5. Horizontal distance from obstruction to receiver: **0.0** ft
6. Horizontal distance from unit to obstruction: **0.0** ft

Detailed Acoustics Information

Octave Band Center Frequency, Hz	31	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	84	63	70	74	78	79	76	71	61	87
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	44	37	54	66	75	79	77	72	60	83
Sound Press. Levels at Dist. Specified above (Lp), dB	59	39	45	50	54	55	52	46	37	62
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	20	12	29	41	50	55	53	47	36	58

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Acoustic Note:

1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. The acoustic center of the unit is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

Acoustic Note:

1. Estimated Sound Power levels - dB re: 1 picowatt
2. Estimated Sound Pressure levels - dB re: 20 micropascal
3. Estimated sound levels given above are assumed to originate at the acoustic center of the unit. The acoustic center of the unit is located at the projection of the condensing unit's geometric center of its base.

4. Sound power levels shown above were determined in accordance with ARI standard 370 for large outdoor refrigeration and air conditioning equipment.

5. Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the ARI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Acoustic Summary For 38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:00AM

Indoor Unit Parameters:

Tag Name:**38AUDT12A0M1-0A0A0 + 40RFAA12A2A1-UA0A0**

Unit Model:..... **40RFA**
Unit Size:..... **10 Tons**
System Type:..... **Dx Cooling Only**
External Static Pressure:..... **0.60** in wg
Fan Speed:..... **1271** RPM
Fan BHP:..... **0.72** BHP

Detailed Acoustics Information

Octave Band Center Frequency, Hz	31	63	125	250	500	1k	2k	4k	8k	Total
Sound Power,dB	NA	99	95	91	90	86	84	80	NA	101
A-Weighted Sound Power, dBA	NA	72	79	82	86	86	85	81	NA	92

Acoustic Notes:

1. 40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
2. The acoustic center of the units is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

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UNIT	ELECTRICAL CHARACTERISTICS	STD. UNIT WT.		CORNER A		CORNER B		CORNER C		CORNER D		CENTER OF GRAVITY			UNIT HEIGHT
		LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	X	Y	Z	H
38AUZ-07 (RTPF)	208/230-3-60,460-3-60,575-3-60	389	176	141	64	96	44	62	28	91	41	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ(A,B)08 (RTPF)	208/230-3-60,460-3-60,575-3-60	391	177	142	64	96	44	62	28	91	41	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ(D,E)08 (RTPF)	208/230-3-60,460-3-60,575-3-60	430	195	142	64	96	44	76	34	111	50	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	490	222	177	80	120	54	78	35	114	52	18 [457.2]	24 [609.6]	24 [609.6]	50-3/8 [1279.2]
38AUZ-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	598	271	195	88	142	64	110	50	151	68	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AUD-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	516	234	185	84	117	53	83	38	131	59	19 [482.6]	23 [584.2]	24 [609.6]	50-3/8 [1279.2]
38AUD-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	654	297	214	97	155	70	120	54	165	75	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AUQ-07	208/230-3-60,460-3-60,575-3-60	444	201	134	61	97	44	90	41	123	56	22 [558.8]	25 [635.0]	13 [330.2]	42-3/8 [1076.0]
38AUQ(A,B)08	208/230-3-60,460-3-60,575-3-60	483	219	162	74	110	50	85	39	125	57	20 [508.0]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUQ(D,E)08	208/230-3-60,460-3-60,575-3-60	523	237	174	79	118	54	96	44	135	61	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]
38AUQ-12	208/230-3-60,460-3-60,575-3-60	575	261	186	84	126	57	106	48	157	71	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]

NOTES:

1. MINIMUM CLEARANCE (LOCAL CODES OR JURISDICTION MAY PREVAIL):
   A. BOTTOM TO COMBUSTIBLE SURFACES: 0 INCHES.
   B. OUTDOOR COIL, FOR PROPER AIR FLOW: 36 INCHES ONE SIDE, 12 INCHES THE OTHER. THE SIDE GETTING THE GREATER CLEARANCE IS OPTIONAL. STANDARD CLEARANCES ON REMAINING TWO SIDES.
   C. OVERHEAD: 60 INCHES, TO ASSURE PROPER OUTDOOR FAN OPERATION.
   D. BETWEEN UNITS: CONTROL BOX SIDE, 42 INCHES PER NEC.
   E. BETWEEN UNIT AND UNGROUNDED SURFACES: CONTROL BOX SIDE, 36 INCHES PER NEC.
   F. BETWEEN UNIT AND BLOCK OR CONCRETE WALLS AND OTHER GROUNDED SURFACES: CONTROL BOX SIDE, 42 INCHES PER NEC.
2. WITH EXCEPTION OF THE CLEARANCE FOR THE OUTDOOR COIL AS STATED IN NOTE 1B, A REMOVABLE FENCE OR BARRICADE REQUIRES NO CLEARANCE.
3. UNITS MAY BE INSTALLED ON COMBUSTIBLE FLOORS MADE FROM WOOD OR CLASS A, B OR C ROOF COVERING MATERIAL.

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES	CONDENSING UNIT		REV
U.S. ECCN:NSR	1 OF 2	06/29/21	05/08/19	38AUZ/ 38AUD/ 38AUQ-07/08/12/14	38AU500349	J

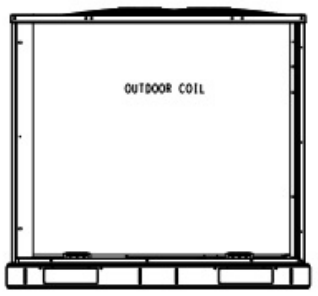
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CG CENTER OF GRAVITY

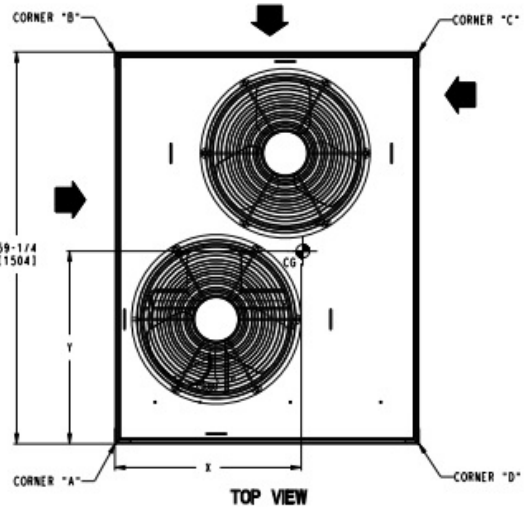
DIRECTION OF AIR FLOW

DIMENSIONS IN [ ] ARE IN MM

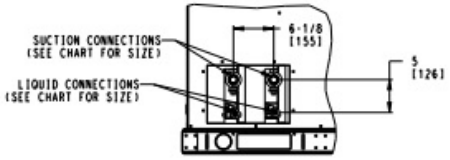
SERVICE VALVE CONNECTIONS			QTY
UNIT	SUCTION	LIQUID	
38AUZ07	1-1/8 [28.6]	3/8 [9.5]	1
38AUZ08	1-1/8 [28.6]	1/2 [12.7]	1
38AUZ12	1-3/8 [34.9]	1/2 [12.7]	1
38AUZ14	1-3/8 [34.9]	5/8 [15.9]	1
38AUD12	1-1/8 [28.6]	3/8 [9.5]	2
38AUD14	1-3/8 [34.9]	1/2 [12.7]	2
38AUQ07	1-1/8 [28.6]	3/8 [9.5]	1
38AUQ08	1-1/8 [28.6]	1/2 [12.7]	1
38AUQ12	1-3/8 [34.9]	1/2 [12.7]	1



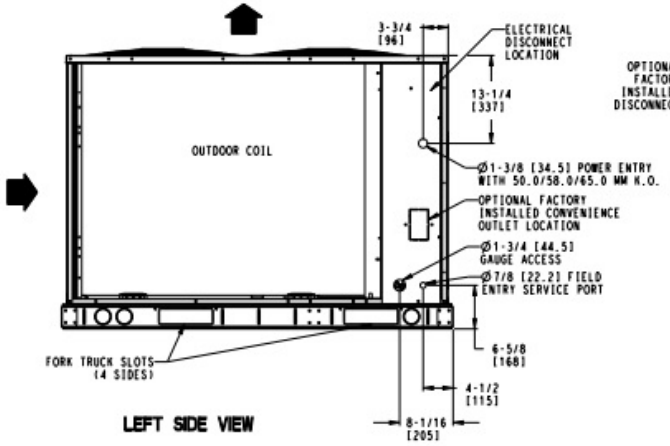
REAR VIEW



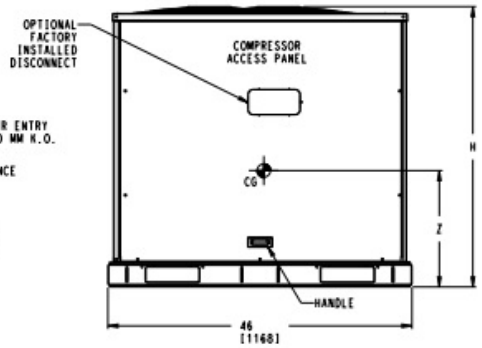
TOP VIEW



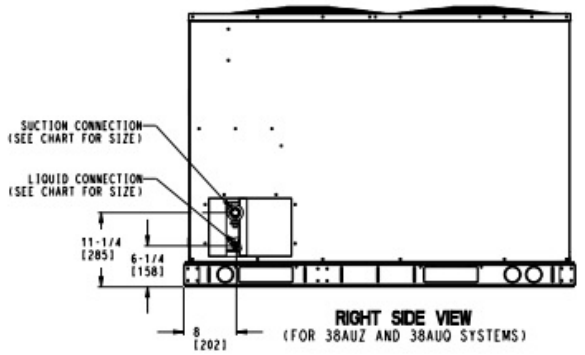
RIGHT SIDE VIEW
(FOR 38AUD SYSTEMS)



LEFT SIDE VIEW

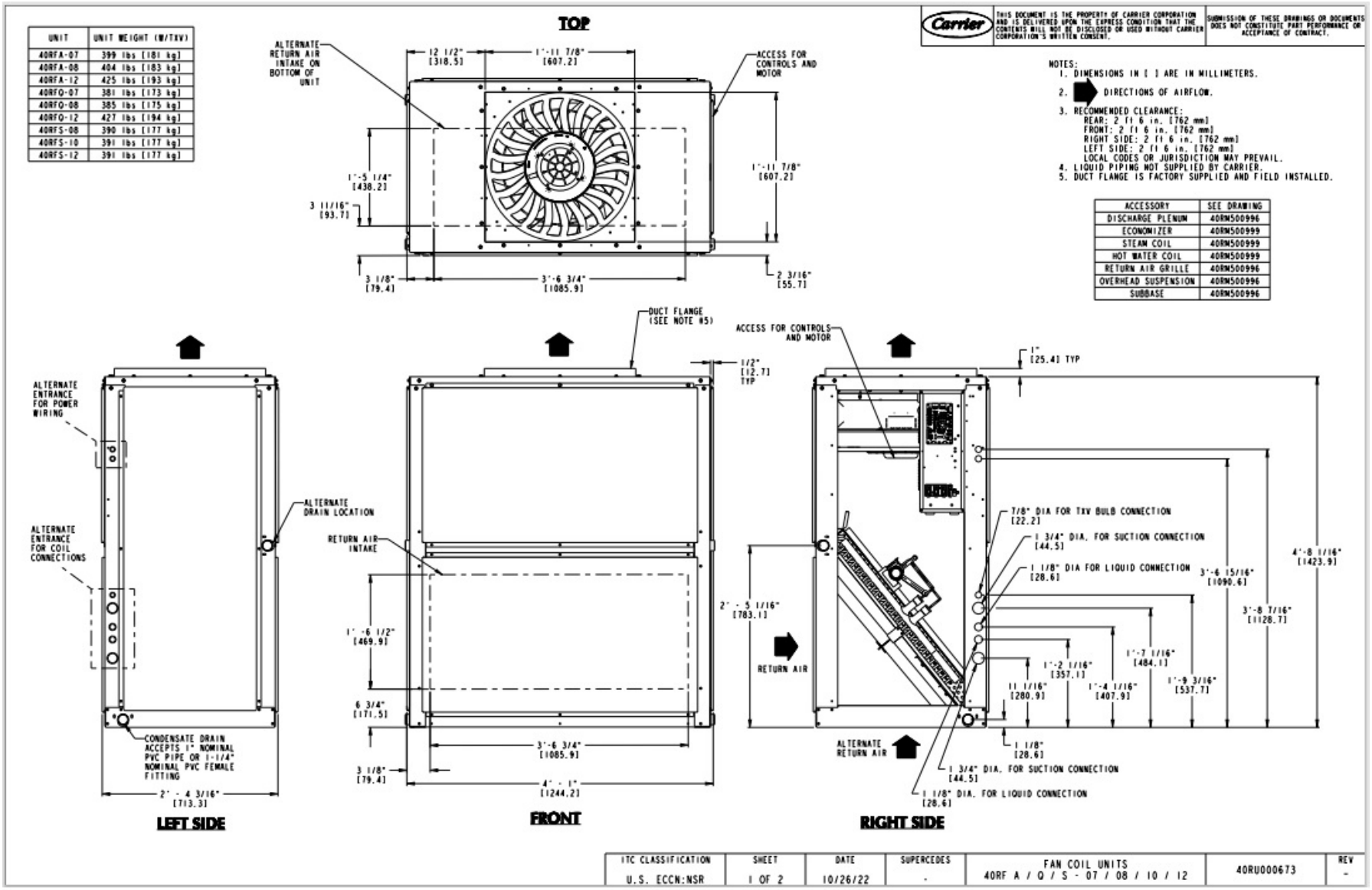


FRONT VIEW



RIGHT SIDE VIEW
(FOR 38AUZ AND 38AUQ SYSTEMS)

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES	CONDENSING UNIT		REV
U.S. ECCN:NSR	2 OF 2	06/29/21	05/08/19	38AUZ/ 38AUD/ 38AUQ-07/08/12/14	38AU500349	J



UNIT	UNIT WEIGHT (W/TXV)
40RFA-07	399 lbs [181 kg]
40RFA-08	404 lbs [183 kg]
40RFA-12	425 lbs [193 kg]
40RFQ-07	381 lbs [173 kg]
40RFQ-08	385 lbs [175 kg]
40RFQ-12	427 lbs [194 kg]
40RFS-08	390 lbs [177 kg]
40RFS-10	391 lbs [177 kg]
40RFS-12	391 lbs [177 kg]

ACCESSORY	SEE DRAWING
DISCHARGE PLENUM	40RM500996
ECONOMIZER	40RM500999
STEAM COIL	40RM500999
HOT WATER COIL	40RM500999
RETURN AIR GRILLE	40RM500996
OVERHEAD SUSPENSION	40RM500996
SUBBASE	40RM500996

GUIDE SPECIFICATIONS – 38AUDT12A0M1-0A0A0

Commercial Air-Cooled Condensing Units
HVAC Guide Specifications

Size: 12

Part 1: General

SYSTEM DESCRIPTION

1.01. Outdoor-mounted, air-cooled condensing unit suitable for on-the-ground or rooftop installation. Unit shall consist of a hermetic scroll air-conditioning compressor(s) assembly, an air-cooled coil, propeller-type condenser fans, and a control box. Unit shall discharge supply air upward as shown on contract drawings. Unit shall be used in a refrigeration circuit matched with a packaged air-handling unit.

QUALITY ASSURANCE

1.01. Unit shall be rated in accordance with AHRI Standard 360.
1.02. Unit construction shall comply with ANSI/ASHRAE 15 safety code latest revision and comply with NEC.
1.03. Unit shall be constructed in accordance with UL 1995 standard and shall carry the UL and UL, Canada label.
1.04. Unit cabinet shall be capable of withstanding 500-hour salt spray exposure per ASTM B117 (scribed specimen).
1.05. Air-cooled condenser coils for hermetic scroll compressor units (38AUZ) and 38AUD shall be leak tested at 150 psig, and pressure tested at 650 psig.
1.06. Unit shall be manufactured in a facility registered to ISO 9001:2000 manufacturing quality standard.

DELIVERY, STORAGE, AND HANDLING

1.01. Unit shall be shipped as single package only, and shall be stored and handled according to unit manufacturer's recommendations.

WARRANTY (FOR INCLUSION BY SPECIFYING ENGINEER.)

Part 2: Products

EQUIPMENT

2.01. General:
  A. Factory-assembled, single piece, air-cooled condensing unit. Contained within the unit enclosure shall be all factory wiring, piping, controls, compressor, holding charge, and special features required prior to field start-up.
2.02. Unit Cabinet:
  A. Unit cabinet shall be constructed of galvanized steel, bonderized and coated with a prepainted baked enamel finish.
  B. A heavy-gauge roll-formed perimeter base rail with forklift slots and lifting holes shall be provided to facilitate rigging.
2.03. Condenser Fans:
  A. Condenser fans shall be direct driven, propeller type, discharging air vertically upward.
  B. Fan blades shall be balanced.
  C. Condenser fan discharge openings shall be equipped with PVC-coated steel wire safety guards.
  D. Condenser fan and motor shaft shall be corrosion resistant.
2.04. Compressor:
  A. Compressor shall be of the hermetic scroll type .

B. Compressor shall be mounted on rubber grommets.
C. Compressors shall include overload protection.
D. Compressors shall be equipped with a crankcase heater.
E. Compressor shall be equipped with internal high pressure and high temperature protection.
F. 38AUZ*16 and 25 sizes shall use two scroll compressors manifold together.

2.05. Condenser Coils:
A. Standard Aluminum fin - Copper Tube Coils:
1. Standard evaporator and condenser coils shall have aluminum lanced plate fins mechanically bonded to seamless internally grooved copper tubes with all joints brazed.
2. Evaporator coils shall be leak tested to 150 psig, pressure tested to 450 psig, and qualified to UL 1995 burst test at 1775 psig.
3. Condenser coils shall be leak tested to 150 psig, pressure tested to 650 psig, and qualified to UL 1995 burst test at 1980 psig.

2.06. Refrigeration Components:
A. Refrigeration circuit components shall include liquid line service valve, suction line service valve, a full charge of compressor oil, and a partial holding charge of refrigerant.

2.07. Controls and Safeties:
A. Minimum control functions shall include:
1. Control wire terminal blocks.
2. Compressor lockout on auto-reset safety until reset from thermostat.
3. Each unit shall utilize the Comfort Alertt Diagnostic Board that provides:
a. System Pressure Trip fault code indication
b. Short Cycling fault code indication
c. Locked Rotor fault code indication
d. Open Circuit fault code indication
e. Reverse Phase 3 fault code indication
f. Welded Contactor fault code indication
g. Low Voltage fault code indication
h. Anti-short cycle protection
i. Phase reversal protection
B. Minimum safety devices which are equipped with automatic reset (after resetting first at thermostat), shall include:
1. High discharge pressure cutout.
2. Low pressure cutout.

2.08. Operating Characteristics:
A. The capacity of the condensing unit shall meet or exceed _____ Btuh at a suction temperature of _____F. The power consumption at full load shall not exceed _____ kW.
B. The combination of the condensing unit and the evaporator or fan coil unit shall have a total net cooling capacity of _____ Btuh or greater at conditions of _____ cfm entering-air temperature at the evaporator at _____F wet bulb and _____F dry bulb, and air entering the condensing unit at _____F.
C. The system shall have an EER of _____ Btuh/Watt or greater at standard AHRI conditions.
D. Standard unit shall be capable to operate up to 125_F (52_C) and down to 40_F (4_C)

2.09. Electrical Requirements:
A. Nominal unit electrical characteristics shall be _____ v, 3-ph, 60 Hz. The unit shall be capable of satisfactory operation within voltage limits of ____ v to _____ v.
B. Unit electrical power shall be single-point connection.
C. Unit control circuit shall contain a 24-v transformer for unit control.

2.10. Special Features:
A. Thermostat Controls:

1. Programmable multi-stage thermostat shall have 7-day clock, holiday scheduling, large backlit display, remote sensor capability, and Title 24 compliance.
2. Commercial Electronic Thermostat shall have 7-day time clock, auto-changeover, multi-stage capability, and large LCD (liquid crystal display) temperature display.

B. Louvered hail Guard Package:
1. Louvered hail guard package shall protect coils against damage from hail and other flying debris.



GUIDE SPECIFICATIONS – 40RFAA12A2A1-UA0A0

Commercial Packaged Air-Handling Unit

HVAC Guide Specifications

Size: 12

Part 1: GENERAL

SYSTEM DESCRIPTION

1.01. Indoor, packaged air-handling unit for use in commercial split systems. Unit shall have a multi-position design and shall be capable of horizontal or vertical installation on a floor or in a ceiling, with or without ductwork. (Only vertical units are to be applied without ductwork.)

1.02. Unit with direct-expansion coil shall be used in a refrigerant circuit with a matching air-cooled condensing unit. Unit with chilled water coil shall be used in a chilled water circuit.

QUALITY ASSURANCE

1.01. Coils shall be designed and tested in accordance with ASHRAE 15 Safety Code for Mechanical Refrigeration (U.S.A.), latest edition.

1.02. Unit shall be constructed in accordance with ETL (U.S.A.) and ETL, Canada, standards and shall carry the ETL and ETL, Canada, labels.

1.03. Unit insulation and adhesive shall comply with NFPA-90A (U.S.A.) requirements for flame spread and smoke generation. Insulation shall contain an EPA-registered immobilized antimicrobial agent to effectively resist the growth of bacteria and fungi as proven by tests in accordance with ASTM standards G21 and 22 (U.S.A.).

1.04. Unit shall be manufactured in a facility registered to the ISO 9001:2000 manufacturing quality standard.

1.05. Direct-expansion and chilled water coils shall be burst and leak tested at 435 psi (2999 kPa).

DELIVERY AND STORAGE

1.01. Units shall be stored and handled per manufacturer's recommendations.

Part 2: Products

EQUIPMENT

2.01. Indoor mounted, draw-thru, packaged air-handling unit that can be used in a suspended horizontal configuration or a vertical configuration. Unit shall consist of a direct drive vane axial fan and motor assembly, pre-wired fan motor controller, factory-installed refrigerant metering devices (direct-expansion coil units), cooling coil, 2 in. (51 mm) disposable air filters, and condensate drain pans for vertical or horizontal configurations.

2.02. Base Unit:

A. Cabinet shall be constructed of mill-galvanized steel.

B. Cabinet panels shall be fully insulated with 1/2-in. (12.7-mm) fire-retardant material. Insulation shall contain an EPA-

registered immobilized antimicrobial agent to effectively resist the growth of bacteria and fungi as proven by tests in accordance with ASTM standards G21 and 22 (U.S.A.).

C. Unit shall contain non-corroding condensate drain pans for both vertical and horizontal applications. Drain pans shall have connections on right and left sides of unit to facilitate field connection. Drain pans shall have the ability to be sloped toward the right or left side of the unit to prevent standing water from accumulating in pans.

D. Unit shall have factory-supplied 2-in. (51 mm) throwaway-type filters installed upstream from the cooling coil. Filter access shall be from either the right or left side of the unit.

2.03. Coils:

A. DX coil is 4-row and consists of copper tubes with sine-wave aluminum fins bonded to the tubes by mechanical expansion. Suction and liquid line connections or supply and discharge connections shall be made on the same side of the coil.

1. Direct-expansion coils shall feature factory installed thermostatic expansion valves (TXVs) for refrigerant control. The TXVs shall be Puron® R-410A compatible and capable of external adjustment. Direct-expansion heat pump coils shall have a factory-installed bypass line and check valve assembly around the TXVs to allow liquid flow from the coil to the outdoor unit during the heating mode. Coil tubing shall be internally rifled to maximize heat transfer.
2. Chilled water coils shall be rated for an operating pressure of not less than 300 psig (2069 kPag).

2.04. Operating Characteristics:

A. Unit shall be capable of providing ____ cfm (L/s) airflow at an external static pressure of ____ in. wg (kPag).

2.05. Motor:

A. Fan motor of the size and electrical characteristics specified on the equipment schedule shall be factory supplied and installed.

B. Motors rated at 1.3 through 3.7 hp (0.97 through 2.76 kW) shall have internal thermal overload protection. Motors rated at 5 hp (3.73 kW) shall be protected by a circuit breaker.

C. Evaporator-fan motor shall have permanently lubricated, sealed bearings and inherent automatic-reset thermal overload protection or manual reset calibrated circuit breakers. Evaporator motors are designed specifically for Carrier and do not have conventional horsepower (hp) ratings listed on the motor nameplate. Motors are designed and qualified in the "air-over" location downstream of the cooling coil and carry a maximum continuous bhp rating that is the maximum application bhp rating for the motor; no "safety factors" above that rating may be applied.

D. All evaporator-fan motors 5 hp and larger shall meet the minimum efficiency requirements as established by the Energy Policy Act of 1992 (EPACT), effective October 24, 1997.

2.06. Special Features:

A. Alternate Motor and Drive:

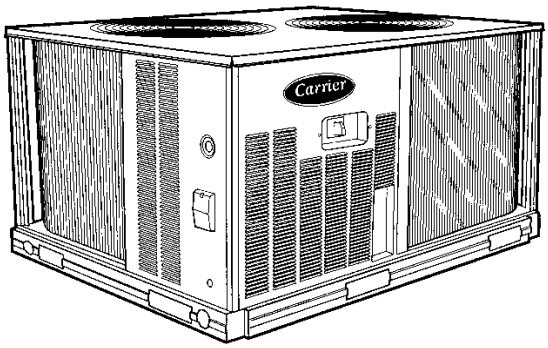
1. An alternate motor and/or medium-static or high-static drive shall be available to meet the airflow and external static pressure requirements specified on the equipment schedule.

B. Thermostat Controls:

1. TEMP System programmable communicating multi-stage thermostat with fan switch, time clock, LCD display, °F/°C capability, and CCN (Carrier Comfort Network™) compatibility.
2. Commercial Electronic Thermostat with 7-day time clock, auto-changeover, multi-stage capability, and large LCD temperature display.

COMMERCIAL SPLIT SYSTEMS
38AUZ 6 to 20 Ton
38AUD 10 to 25 Ton
AIR-COOLED COOLING ONLY CONDENSING UNITS with PURON® REFRIGERANT

These light commercial air-cooled cooling only condensing units are easily connected by refrigerant lines and low voltage control wiring to matching Carrier packaged air-handling units or other suitable evaporator units. They are ideal for new construction or renovation applications where quality and performance are required.



ASHRAE 90.1 COMPLIANT

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AHRI CERTIFIED www.ahridirectory.org Unitary Large AC AHRI Standard 340/360

Gemini

Puron the environmentally sound refrigerant

Manufactured in facility registered to ISO 9001:2008 quality standard

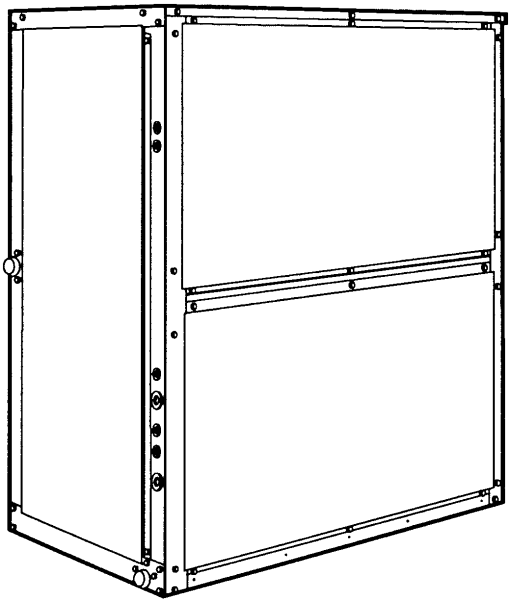
BASE UNIT STANDARD FEATURES:

- Puron® (R-410A) HFC refrigerant partial charge
- Single circuit units (38AUZ) are available in 2 stage cooling
- Two circuit units (38AUD) are available in 3 stage cooling
- All models utilize round tube plate fin coils.
- Standard one year parts warranty and five year compressor parts warranty. Extended 2-5 year parts and 1-5 labor coverage are available.
- Cooling operating range up to 125°F (52°C) and down to 35°F (2°C) as standard.
- Available Motormaster® I control for cooling operation down to -20°F (-29°C)
- Fully hermetic scroll compressor with crankcase heater
- New terminal board facilitating simple safety circuit troubleshooting and simplified control box arrangement. 24-volt 75va control system
- High and low pressure switches.

- Comfort Alert™ Diagnostic Board:
  - LED Go-No-Go and fault code
  - Built in time guard anti-short cycle
  - Phase protection
  - Fault code retention logic
  - Low voltage compressor contactor protector
- UL and UL, Canada apply to standard units; 575-volt units UL, Canada only.
- Full perimeter base rails with built-in rigging adapters and fork truck slots
- Pre-painted exterior panels and primer-coated interior panels tested to 500 hours salt spray protection
- Direct drive permanently lubricated condenser fan motors
- All units factory run tested
- Compressors mounted on independent vibration isolators

COMMERCIAL SPLIT SYSTEMS
40RFA 6 to 10 ton
PACKAGED AIR HANDLING UNITS with PURON® REFRIGERANT

The 40RFA units are completely packaged direct expansion air-handling units offering the ultimate in installation ease and flexibility. Combined with Carrier 38A Series condensing units these packaged air handling units offer system capacities from 6 up to 10 tons. These units may be installed in either the vertical or hoizontal configuration with no modifications. Powerful fan systems allow easy adaptation to existing ductwork. Heating capability is provided by electric heaters, hot water coils or steam coils.



Manufactured in facility registered to
ISO 9001:2008 quality standard

BASE UNIT STANDARD FEATURES:

- High efficiency Ecoblue Van Axial Fan System
- Unit Control Board (UCB) standard to allow for easy set up
- Puron® (R-410A) HFC refrigerant compatible
- 6 and 7.5 ton units feature a single evaporator coil, 10 ton units feature two evaporator coils.
- Designed for vertical or horizontal installation
- Standard sloped drain pans and cleanable insulation
- All models utilize round tube plate fin coils for enhanced cooling performance
- Standard one year parts warranty. Extended 2-5 year parts and 1-5 labor coverage are available.
- High-static design meets a wide range of applications
- Standard factory installed thermostatic expansion valves (TXV) with removable power help to reduce installation time
- Die-formed galvanized steel casings provide durability and structural integrity. Optional paint is available. Optional Staged Air volume (SAV™) system provides two-speed fan control for energy savings
- Ultra LOW LEAK Economizer accessory provides ventilation air and “free” cooling with built in Fault Detection and Diagnostic (FDD) capabilities.
- Hot water coil, steam coil, and electric heat accessories are available.